

# Carl A. Schmidt

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## Appointments

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|-------------------------------------------------------------------------|----------------|
| Research Assistant Professor, Boston University                         | 2021 - Present |
| Research Scientist, Boston University                                   | 2017 - 2021    |
| Research Associate, Sorbonne Univ. CNRS, Paris (F. Leblanc Supervisor)  | 2015 - 2017    |
| Research Associate, Univ. of Virginia (R. E. Johnson Supervisor)        | 2013 - 2015    |
| Graduate Research Assistant, Boston Univ. (M. Mendillo Supervisor)      | 2006 - 2013    |
| Undergraduate Research Assistant, Univ. Colorado (F. Hearty Supervisor) | 2002 - 2005    |

**Expertise:** Planetary aeronomy, Monte-Carlo modelling, telescope-based spectroscopy & instrumentation, aurorae & plasmas

## Education

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|-----------------------------------------------------------------|------|
| Ph.D., Astronomy, Boston University                             | 2013 |
| Thesis: <i>Mercury's Sodium Exosphere</i> (M. Mendillo Advisor) |      |
| B.A., Physics, University of Colorado                           | 2005 |

## Advising

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Post-Doctoral and Research Scientists:

- Yudai Suzuki (Fall 2025 -) JSPS Postdoctoral Fellow
- Candace Gray (Fall 2025 -) Research Scientist
- Quanzhi Ye (Fall 2022 -) Visiting Researcher (Joint appt w/ Univ. of Maryland)

Graduate Research Advisor:

- Patrick Lierle (Fall 2022 -)
- Emma Lovett (Fall 2021 -) NASA FINESST Graduate Fellow

Undergraduate Research Advisor:

- Madison Jordan (Summer 2024) North Carolina A&T, NSF PAARE internship
- Aishwarya Ganesh (Summer 2021) Univ. Texas, NASA SUPPR internship
- Cameron Moye (Summer 2020) Univ. Maryland, NASA SUPPR internship

- Patrick Lierle (Summer 2019 - Summer 2022)
- Mikhail Sharov (Fall 2018 - Spring 2021)
- Chase Young (Spring - Summer 2018)

## Peer-Reviewed Journal Articles

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53 articles, students underlined.

- Z. Milby, K. de Kleer, **C. Schmidt** (2026) *Variability of Callisto's Optical Aurora in Eclipse*, The Planetary Science Journal, *in press*
- A. V. Oza, R. E. Johnson, **C. A. Schmidt**, W. M. Calvin (2026) *Plasma and Thermal Processing Leading to Spatial and Temporal Variability of the Trapped O<sub>2</sub> at Europa and Ganymede*, Astrobiology, *in press*
- **C. Schmidt**, L. Roth, S. Mendenhall, K. Retherford, J. Spencer (2026) *A Lack of Carbon at Io and its Plasma Torus*, Icarus, V454, 117088, [DOI](#).
- Z. Milby, K. de Kleer, **C. Schmidt**, M. Camarca (2026) *Variability of Europa's Optical Aurora on Orbital and Multiyear Timescales*. The Planetary Science Journal, V7, 60, [DOI](#).
- K. L. Knowles, H. Melin, T. S. Stallard, L. Moore, J. O'Donoghue, **C. Schmidt**, J. R. Szalay, P. I. Tiranti, K. Roberts, R. E. Johnson, E. M. Thomas (2026) *Short-Term Variability of Jupiter's Satellite Footprints with JWST*, Geophysical Research Letters, V53 (5), e2025GL118553, [DOI](#).
- K. Roberts, L. Moore, J. O'Donoghue, H. Melin, T. S. Stallard, I. Muller-Wodarg, O. Agiwal, K. L. Knowles, P. I. Tiranti, K. Mohamed, **C. Schmidt** (2026) *A Global View of Jupiter's Upper Atmosphere Through H<sub>3</sub><sup>+</sup>*, Astrophysical Journal Letters, V998, L13, [DOI](#).
- L. Moore, H. Melin, T. S. Stallard, J. O'Donoghue, K. Roberts, P. I. Tiranti, K. Mohamed, O. Agiwal, I. Muller-Wodarg, K. L. Knowles, P. Withers, D. Coffin, R. Wang, **C. Schmidt** (2025) *Photochemistry in Jupiter's Ionosphere: Insights from Simultaneous H<sub>3</sub><sup>+</sup> and Electron Density Observations During Juno Perijove 54*. The Planetary Science Journal, V7 (1), 25, [DOI](#).
- L. Naesenius, L. Roth, N. Ivchenko, S. Carberry Mogan, P. M. Molyneux, K. D. Retherford, J. Saur, **C. Schmidt**, S. Bergman (2026) *Lyman- $\alpha$  observations of Ganymede emerging from eclipse reveal a stable hydrogen exosphere*. Monthly Notices of the Royal Astronomical Society, V545 (4), staf2204, [DOI](#).
- I. de Pater, D. F. Strobel, A. G. Davies, J. Saur, L. Roth, K. de Kleer, E. Lellouch, Z. Milby, **C. Schmidt**, T. Fouchet, M. H. Wong, L. N. Fletcher, J. Harkett, M. Roman, R. J. Cartwright, J. E. Perry, J. Renaud-Kim, C. Jordan, S. Schlegel, D. A. Williams, D. M. Nelson (2025) *First Detection of [S I] in Near-IR JWST Observations of Io in Eclipse, and Comparison with SO Emissions, Evolving Volcanic Eruptions, and prior UV HST-STIS [S I] emissions*. Journal of Geophysical Research: Planets, V130(11), e2024JE008850, [DOI](#).

- A. V. Oza, A. Gebek, M. Meyer zu Westram, A. Tokadjian, A. Piro, R. Hu, A. Unni, R. Chari, Aaron Bello-Arufe, **C. Schmidt**, A. Louca, Y. Miguel, R. Estrela, J. Yang, M. Damiano, Y. Hasegawa, L. Welbanks, D. Powell, Y. K. Yung (2025) *Volcanic Satellites Tidally Venting Na, K, SO<sub>2</sub> in Optical & Infrared Light*, Monthly Notices of the Royal Astronomical Society, staf1526, [DOI](#).
- [P. R. Lierle](#), [E. Lovett](#), **C. Schmidt**, A. Merkel (2025) *Characterizing Dynamical Processes in Surface-Bound Exospheres Via Resolved Sodium D Emissions*, Frontiers in Astronomy and Space Sciences, V12, 1585683, [DOI](#).
- Z. Milby, K. de Kleer, **C. Schmidt** (2025) *Detection of New Auroral Emissions at Io and Implications for Its Interaction with the Plasma Torus*, The Planetary Science Journal, V6, 181, [DOI](#).
- [E. Lovett](#), **C. Schmidt**, [P. R. Lierle](#) (2025) *Europa's Sodium and Potassium Exosphere during Juno's Flyby*, The Planetary Science Journal, V6, 178, [DOI](#).
- [P. R. Lierle](#) & **C. Schmidt** (2025) *Sodium and Potassium Linewidths as an Atmospheric Escape Diagnostic at Mercury*, Journal of Geophysical Research: Planets, V130 (7), e2025JE008975, [DOI](#).
- C. Nixon and 38 co-authors, including **C. Schmidt** (2025) *Titan's Atmosphere in Late Northern Summer from JWST and Keck Observations*. Nature Astronomy, V9, pp. 969-981, [DOI](#).
- A. Unni, A. V. Oza, S. Thirupathi, H. J. Hoeijmakers, J. V. Seidel, **C. A. Schmidt**, K. de Kleer, A. D. Baker, B. Manjunath, A. Gebek, M. Meyer zu Westram (2025) *Doppler Shifted Transient Sodium Detection by KECK/HIRES*, Monthly Notices of the Royal Astronomical Society: Letters, V540, L48–L53, [DOI](#).
- K. Roberts, L. Moore, J. O'Donoghue, H. Melin, T. Stallard, K. Knowles, **C. Schmidt**, P. Tiranti (2025) *Spatiotemporal Variations of Temperature in Jupiter's Upper Atmosphere*. The Planetary Science Journal, V6, 92, [DOI](#).
- L. Roth, A. Blöcker, K. de Kleer, D. Goldstein, E. Lellouch, J. Saur, **C. Schmidt**, D.F. Strobel, C. Tao, F. Tsuchiya, V. Dols, H. Huybrighs, A. Mura, J. R. Szalay, S. V. Badman, I. de Pater, A.-C. Dott, M. Kagitani, L. Klaiber, R. Koga, A. McEwen, Z. Milby, K.D. Retherford, S. Schlegel, N. Thomas, W.L. Tseng, A. Vorburger (2025) *Mass Supply from Io to Jupiter's magnetosphere*. Space Science Reviews, 221, 13, [arXiv](#), [DOI](#).
- C. Gray, K. Peter, M. Pätzold, S. Tellmann, T. Nordheim, **C. Schmidt**, N. Chanover and P. Withers (2025) *Venus' O 5577Å Oxygen Green Line: A Global Diffuse Proton-induced Aurora*. Journal of Geophysical Research: Space Physics, V130 (2), e2024JA032851, [DOI](#).
- E. G. Nerney, F. Bagenal, **C. Schmidt** (2024) *Simulations of Optical Emissions in Io's Plasma Torus*. Journal of Geophysical Research: Space Physics, V130 (1), e2024JA033232, [DOI](#).
- **C. Schmidt** (2024) *On the Implications of Ground-Based High-Definition Imaging of Io's Surface*. Geophysical Research Letters, V51, 21, e2024GL112169, [DOI](#).

- A. V. Oza, J. V. Seidel, H. J. Hoeijmakers, A. Unni, A. Y. Kesseli, **C. Schmidt**, S. Thirupathi, A. Gebek, M. Meyer zu Westram, S. Sousa, A. Bello-Arufe, R. Lopes, R. Hu, C. Fisher, S. Charnoz, A. D. Baker, S. P. Halverson, K. de Kleer, N. Schneider, A. Psaridi, M. Lendl, A. Wyttenbach, I. Bhatnagar and R. E. Johnson (2024) *Redshifted Sodium Transient Near Exoplanet Transit*, The Astrophysical Journal Letters, 973, L53, [DOI](#).
- M. Yoneda, F. Tsuchiya, **C. Schmidt**, M. Kagitani and T. Sakanoi (2024) *Major Brightening Events in Jupiter’s Sodium Nebula during Juno era*. Icarus, V425, [DOI](#).
- Z. Milby, K. de Kleer, **C. Schmidt**, and F. Leblanc (2024) *Short-Timescale Spatial Variability of Ganymede’s Optical Aurora*, The Planetary Science Journal, V5, 153, [DOI](#).
- J. Morgenthaler, J. Rathbun, **C. Schmidt**, J. Baumgardner and N. Schneider (2024) *Erratum: ‘Large Volcanic Event on Io Inferred from Jovian Sodium Nebula Brightening’ (ApJL 871:L23)*, The Astrophysical Journal Letters, 966 (2), L40, [DOI](#).
- J. Morgenthaler, **C. Schmidt**, M. F. Vogt, N. M. Schneider and M. Marconi (2024) *Jovian Sodium Nebula and Io plasma torus  $S^+$  and Brightnesses 2017 – 2023: Insights into Volcanic vs. Sublimation Supply*. Journal of Geophysical Research: Space Physics, 129, 3, [DOI](#).
- D. Bhattacharyya, J. T. Clarke, M. Mayyasi, V. Shematovich, D. Bisikalo, J. Y. Chaufray, E. Thiemann, J. Halekas, **C. Schmidt**, J. L. Bertaux, M. S. Chaffin, and N. M. Schneider (2023) *Evidence of Hot Hydrogen in the Exosphere of Mars Resulting in Enhanced Water Loss*. Journal of Geophysical Research: Planets, 128, 8, [DOI](#).
- P. Lierle, **C. Schmidt**, J. Baumgardner, L. Moore and E. Lovett (2023) *The Rapid Imaging Planetary Spectrograph*. Publications of the Astronomical Society of the Pacific, 135, 095002, [arXiv](#), [DOI](#).
- Q. Zhang, K. Battams, Q. Ye, M. Knight and **C. Schmidt** (2023) *Sodium Brightening of (3200) Phaethon Near Perihelion*. The Planetary Science Journal, V4, 70, [DOI](#).
- K. de Kleer, Z. Milby, **C. Schmidt**, M. Camarca and M. Brown (2023) *The Optical and Near-Infrared Aurorae of Europa, Ganymede and Callisto*. The Planetary Science Journal, V4, 37, [DOI](#).
- **C. Schmidt**, M. Sharov, K. de Kleer, N. Schneider, I. de Pater, P.H. Phipps, A. Conrad, L. Moore, P. Withers, J. Spencer, J. Morgenthaler, I. Ilyin, K. Strassmeier, C. Veillet, J. Hill, and M. Brown (2023) *Io’s Optical Aurorae in Jupiter’s Shadow*. The Planetary Science Journal, V4, 36, [DOI](#).
- P. R. Lierle, **C. Schmidt**, J. Baumgardner, L. Moore, T. Bida and R. Swindle (2022) *The Spatial Distribution and Temperature of Mercury’s Potassium Exosphere*. The Planetary Science Journal, V3, 87, [DOI](#).
- **C. Schmidt** (2022) *Doppler-Shifted Alkali D Absorption as Indirect Evidence for Exomoons*. Frontiers in Astronomy and Space Sciences, Vol. 9, 801873, [DOI](#).
- A. L. E. Werner, S. Aizawa, F. Leblanc, J. Y. Chaufray, R. Modolo, J. M. Raines, W. Exner, U. Motschmann and **C. Schmidt** (2022) *Ion density and phase space density*

- distribution of planetary ions  $\text{Na}^+$ ,  $\text{O}^+$  and  $\text{He}^+$  in Mercury's magnetosphere.* Icarus, V372, 114734, [DOI](#).
- T. Cassidy, **C. Schmidt**, A. Merkel, J. Jasinski and M. Burger (2021) *Detection of Large Exospheric Enhancements at Mercury due to Meteoroid Impacts*, The Planetary Science Journal, Vol. 2, 175, [DOI](#).
  - J. Baumgardner, S. Luetttgen, **C. Schmidt**, M. Mayyasi, S. Smith, C. Martinis, J. Wroten, L. Moore and M. Mendillo (2021) *Long-Term Observations and Physical Processes in the Moon's Extended Sodium Tail*, Journal of Geophysical Research: Planets, V126, 3, [DOI](#).
  - V. Mangano and 61 co-authors including C. Schmidt (2021) *BepiColombo science investigations during cruise and flybys at the Earth, Venus and Mercury*, Space Science Reviews, V217, 23, [DOI](#).
  - **C. Schmidt**, J. Baumgardner, L. Moore, T. A. Bida, R. Swindle and P. Lierle (2020) *The Rapid Imaging Planetary Spectrograph: Observations of Mercury's Sodium Exosphere in Twilight*. The Planetary Science Journal, V1, 4, [DOI](#).
  - A. Oza, R.E. Johnson, E. Lellouch, **C. Schmidt**, N. Schneider, C. Huang, D. Gamborino, A. Gebek, A. Wyttenbach and B-O Demory. (2019) *Sodium and Potassium as Remnants of Volcanic Satellites Orbiting Close-in Gas Giant Exoplanets*, The Astrophysical Journal, V885, 2, [DOI](#).
  - L. Moore, H. Melin, T. Stallard, J. O'Donoghue, J. Moses, S. Miller and **C. Schmidt** (2019) *Modelling  $\text{H}_3^+$  in Planetary Atmospheres: Effects of Vertical Gradients on Observed Quantities*, Philosophical Transactions of the Royal Society A, V377, 2154, [DOI](#).
  - R.E. Johnson, A. Oza, F. Leblanc, **C. Schmidt** and T.A. Nordheim (2019) *The Origin and Fate of  $\text{O}_2$  in Europa's Ice: An Atmospheric Perspective*. Space Science Reviews, 215 (1), 20, [DOI](#).
  - J. Morgenthaler, J. Rathbun, **C. Schmidt**, J. Baumgardner and N. Schneider (2019) *Large Volcanic Event on Io Inferred from Jovian Sodium Nebula Brightening*, The Astrophysical Journal Letters, 871 (2), L23, [DOI](#).
  - A. Oza, F. Leblanc, R. E. Johnson, **C. Schmidt**, L. Leclercq, T. Cassidy and J.-Y. Chaufray (2019) *Dusk Over Dawn  $\text{O}_2$  Asymmetry in Europa's Near-Surface Atmosphere*. Planetary & Space Science, 167, pp. 23-32, [DOI](#).
  - **C. Schmidt**, N. Schneider, F. Leblanc, C. Gray, J. Morgenthaler, J. Turner and C. Grava (2018) *A Survey of Visible  $\text{S}^+$  Emission in Io's Plasma Torus During the Hisaki Epoch*. Journal of Geophysical Research: Space Physics, 123, 7, pp. 5610-5624, [DOI](#).
  - F. Leblanc, A. Oza, L. Leclercq, **C. Schmidt**, T. Cassidy, R. Modolo, J.Y. Chaufray and R. E. Johnson (2017) *On the Orbital Variability of Ganymede's Atmosphere*. Icarus, Vol. 293, pp. 185-198, [DOI](#).
  - J. D. Turner, D. Christie, P. Arras, R. E. Johnson and **C. Schmidt** (2016) *Investigation of the environment around close-in transiting exoplanets using CLOUDY*. Monthly Notices of the Royal Astronomical Society, V458 (4), pp. 3880-3891, [DOI](#).
  - **C. Schmidt** (2016) *High Resolution Integral-Field Spectroscopy of Gas and Ion Distributions in the Coma of Comet C/2012 S1 ISON*. Icarus, V265, pp. 35-41, [DOI](#).

- R.E. Johnson, A. Oza, L.A. Young, A.N. Volkov and **C. Schmidt** (2015) *Volatile Loss and Classification of Kuiper Belt Objects*. The Astrophysical Journal, V809 (1), 43, [DOI](#).
- N.-E. Raouafi, C. M. Lisse, G. Stenborg, G. H. Jones and **C. Schmidt** (2015) *Dynamics of HVECs emitted from comet C/2011 L4 as observed by STEREO*. Journal of Geophysical Research, V120 (7), pp. 5329-5340, [DOI](#).
- **C. Schmidt**, R.E. Johnson, J. Baumgardner and M. Mendillo (2015) *Observations of Sodium in the Coma of Comet C/2012 S1 (ISON) During Outburst*. Icarus, V247, pp. 313-318, [DOI](#).
- **C. Schmidt** (2013) *Monte-Carlo Modeling of North-South Asymmetries in Mercury's Sodium Exosphere*, Journal of Geophysical Research, V118, A50396, [DOI](#).
- **C. Schmidt**, J. Baumgardner, M. Mendillo and J. Wilson (2012) *Escape rates and variability constraints for high-energy sodium sources at Mercury*, Journal of Geophysical Research, V117, A03301, [DOI](#).
- **C. Schmidt**, J. Wilson, J. Baumgardner and M. Mendillo (2010) *Orbital Effects on Mercury's Escaping Sodium Exosphere*, Icarus, Vol 207 (1), pp. 9-16, [DOI](#).

## Articles Under Review

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- E. Lovett, **C. Schmidt**, Z. Milby, S. Ellis, J. Szalay (2026) *Europa's Neutral Sodium Cloud is Non-Collisional and Stable Over Decades*, Journal of Geophysical Research: Space Physics, *under review*

## Peer-Reviewed Book Chapters

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- **C. Schmidt** and J. Baumgardner (2022) *Lunar Atmosphere, Alkali Lunar Exosphere* in *Encyclopedia of Lunar Science*, Editors B. Cudnik & C. Ahrens, Springer, [DOI](#).
- F. Leblanc, **C. Schmidt**, V. Mangano, A. Mura, G. Cremonese, J. M. Raines, J.M. Jasinski, M. Sarantos, A. Milillo, R.M. Killen, S. Massetti, T. Cassidy, R.J. Vervack Jr., S. Kameda, M.T. Capria, M. Horanyi, D. Janches, A. Berezhnoy, A. Christou, T. Hirai, P. Lierle and J. Morgenthaler (2022) *Comparative Na and K Mercury and Moon exospheres* in *Surface Bounded Exospheres and Interactions in the Solar System*, Space Sciences Series of ISSI, Springer. Jointly published in Space Science Reviews, Vol 218, 2, [DOI](#).

## Non-Peer-Reviewed Publications

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- Zhang, Q., Knight, M., Ye, Q., Schmidt, C., Battams, K. (2026) *Preliminary Nucleus Size Estimate for Kreutz Sungrazer C/2026 A1 (MAPS)*, Research Notes of the AAS, V10, Issue 3, 57, [DOI](#).



- Chanover, N., Schmidt, C., DeColibus, D. (2021) *The Continued Relevance of 4m Class Telescopes to Planetary Science in the 2020s*, White paper #497 submitted to the Decadal Survey in Planetary Science and Astrobiology 2023-2032, Bulletin of the AAS, 53(4), [DOI](#).
- P. Prem, A. Kereszturi, A. Deutsch, C. Hibbitts, C. Schmidt and 36 co-authors (2021) *Lunar Volatiles and Solar System Science*, White paper #68 submitted to the Decadal Survey in Planetary Science and Astrobiology 2023-2032, Bulletin of the AAS, 53(4), [ArXiv](#), [DOI](#).
- A. Deutsch, N. Chabot, A. Maiti, A. Luspai-Kuti, A. Kereszturi, A. Lucchetti, A. Virkki, A. Colaprete, A. Vorburger, B. Byron, B. Jones, B. Anzures, B. Butler, C. Schmidt and 59 co-authors (2021) *Science Opportunities offered by Mercury's Ice-Bearing Polar Deposits*. Whitepaper #69 submitted to the Planetary Science and Astrobiology Decadal Survey 2023-2032, Bulletin of the AAS, 53(4), [DOI](#).
- J. Clarke, C. Schmidt, J. Baumgardner, C. Carveth, M. Matta, M. Mendillo, L. Moore, and P. Withers (2013) White Paper on Comparative Planetary Exospheres. White paper submitted to Solar and Space Physics Decadal Survey, [ref](#), [link](#).
- F. Hearty, S. Beland, J. Green, N. Cunningham, J. Barentine, M. Drosback, R. Valentine, A. Bondarenko, C. Schmidt, J. Walawender, C. Froning, J. Morse, P. Hartigan (2005) Colorado's Near-Infrared Camera (AKA NIC-FPS) Commissioning on the ARC 3.5M Telescope, Proc. SPIE, Vol 5904, p. 199-211, [DOI](#).

## Invited Talks and Colloquia

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| Mercury's Sodium Exosphere, Mercury 2026, Leuven, BE                                     | 2026 |
| Lunar Sodium Exosphere Imaging, Taiwan Lunar Symposium, TW                               | 2026 |
| Intro to the Jupiter System, NASA Community College Network, Virtual                     | 2025 |
| The Io Plasma Torus, Union College, Schenectady, NY, USA                                 | 2024 |
| Io's Atmosphere, Ionosphere, and Plasma Torus, LATMOS, FR                                | 2023 |
| Solar System Context in the Hunt for Exomoons, NASA JPL, Pasadena, CA, USA               | 2023 |
| Alkali Emissions in the Lunar Atmosphere, UMASS, Lowell, MA, USA                         | 2023 |
| The Lunar Sodium Exosphere, Taiwan Space Union, TW                                       | 2023 |
| The Io-Jupiter Interaction, UMD, College Park, MD, USA                                   | 2022 |
| Io's Atmosphere and Plasma Torus, Boise State University, Boise, ID, USA                 | 2022 |
| Optical Spectroscopy of Jupiter's Moons, AAVSO, Cambridge, MA, USA                       | 2021 |
| Observing the Exospheres of Mercury & the Moon, UMASS, Lowell, MA, USA                   | 2020 |
| Io's Escaping Atmosphere & Plasma Torus, Boston College, Boston, MA, USA                 | 2018 |
| Solar Transit Spectroscopy of Mercury's Exosphere, Universiteit van Amsterdam, NL        | 2018 |
| Io's Escaping Atmosphere & Plasma Torus, Universität zu Köln, DE                         | 2018 |
| Io's Volcanic Atmosphere and Plasma Torus, Boston University, Boston, MA, USA            | 2018 |
| Io's Plasma Torus Density & the S <sup>+</sup> Ribbon, Royal Institute of Technology, SE | 2017 |
| Small Telescope Applications: Mercury, Io & Comets, Université de Liège, BE              | 2017 |
| Planetary Applications for Small Telescopes, Institute of Astronomy, Sofia, BG           | 2017 |
| Visible Spectroscopy of the Io Plasma Torus, LESIA, l'Observatoire de Paris, FR          | 2016 |

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| Observations of Io, its Plasma Torus and Neutral Clouds, Lancaster Univ, UK    | 2016 |
| Modern Planetary Applications for Small Telescopes, UMD, College Park, MD, USA | 2015 |
| Characteristics of Sodium Escape at Mercury, SERENA-HEWG, Killarney, IRL       | 2014 |
| Atmospheric Escape in Our Solar System, Space Challenges, Sofia, BG            | 2013 |
| Mercury's Sodium Atmosphere, AOSS, Univ. of Michigan, Ann Arbor, MI, USA       | 2012 |
| Mercury's Tenuous Atmosphere, Heliophysics, NASA GSFC, Greenbelt, MD, USA      | 2012 |

## Contributed Talks and Conference Proceedings

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As of 2026: 27 first author, 84 co-author, 17 student led

## Grant Awards & Fellowships

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27 grants (\$5.0M total, \$3.2M to Boston University)

- NASA/NExSCI Keck Award *Plasma drivers of Io's SO emission via joint Keck and JWST observations*, PI, 2026.02.01 to 2027.01.31. 7/2026A-N076. Total budget / funding to BU: \$10,250.
- James Webb Space Telescope Cycle 4 *Io's Auroral Emissions as a Tool to Investigate Atmosphere-Plasma Torus Interactions*", Co-I (PI Zac Milby, Caltech), 2025.07.01 to 2028.06.30, JWST-GO-08857. Total budget: \$110,475. Funding to BU: \$19,999
- Hubble Space Telescope Cycle 32 *A UV eclipse test for Callisto's atmosphere, plasma interaction and water ocean*, Co-I (PI Kurt Retherford, SwRI), 2025.03.01 to 2028.02.28. HST-GO-17739. Total budget: \$63,963. Funding to BU: \$5,000
- NASA/NExSCI Keck Award *Joint Keck-Juno observations of Jupiter, its moons and its magnetosphere*, PI, 2024.08.01 to 2026.07.30. 80NSSC22K0954 (renewal). Total budget / funding to BU: \$120,000
- Hubble Space Telescope Cycle 31 *HST-Juno Io Campaign: Connecting Volcanos to the Plasma Environment*, Co-I (PI Kurt Retherford, SwRI), 2023.12.01 to 2026.11.30. HST-GO-17470. Total budget: \$268,968. Funding to BU: \$13,160
- NSF AST PAARE *Accessing the dark Arizona skies for research and education, a NCAT-BU partnership*, Co-PI (PI Dan Clemens, BU), 2023.08.24-2026.08.23. Total budget / funding to BU: \$256,696
- NASA/NExSCI Keck Award *Recoil Heating in Mercury's Alkali Exotail*, PI (Science PI Patrick Lierle, BU), 2024.02.01 to 2025.01.31. Total budget / funding to BU: \$13,375.
- NASA FINESST PI for Emma Lovett's Graduate Fellowship *Characterizing and Simulating Alkalies in Europa's Exosphere*, PI, 2023.09.01 to 2026.08.31. 22-PLANET22-0198. Total budget / funding to BU: \$150,000
- Hubble Space Telescope Cycle 30 *Ganymede's water atmosphere in eclipse*, PI, 2023.06.01 to 2026.05.31. HST-GO-17099. Total budget: \$75,414. Funding to BU: \$55,643



- NASA Discovery Data Analysis Program *Mercury's escaping sodium tail*, PI, 2022.09.15 to 2025.09.14. 80NSSC22K1303. Total budget: \$525,000. Funding to BU: \$393,997
- NASA/NEExSCI Keck Award *Joint Keck-Juno observations of Jupiter, its moons and its magnetosphere*, PI, 2022.08.01 to 2024.07.30. 80NSSC22K0954. Total budget / funding to BU: \$150,000
- NSF Astronomy and Astrophysics Research Grant. *Mass transport in Jupiter's magnetosphere: driven by internal or external processes?* Co-I/Institutional PI (PI Jeff Morgenthaler, Planetary Science Institute), 2021.09.01 to 2024.08.30. AST-2108416. Funding to BU: \$94,720
- NASA Solar System Observations *Dynamic Processes on the Galilean Satellites*, Co-I/Institutional PI (PI John Spencer, Southwest Research Institute), 2021.08.01 to 2024.07.31. 80NSSC21K1138. Funding to BU: \$59,981
- NASA Discovery Data Analysis Program *Investigating the Impactor Contribution to Mercury's Exosphere*, Co-I/Institutional PI (PI Aimee Merkel, Univ. Colorado), 2021.05.21 to 2024.04.30. 80NSSC21K1019. Total budget: \$551,469. Funding to BU: \$131,866
- NASA Science Mission Directorate *Characterizing Mercury's Exosphere with BepiColombo-PHEBUS: US-based Co-Investigators*, PI, 2020.10.13 to 2025.10.12, 80NSSC21K0051. Total budget / funding to BU: \$226,061
- NASA/NEExSCI Keck Award *Response of Io's atomic atmosphere and ionosphere to Jovian eclipse: joint observations with HIRES and HST*, PI, 2020.02.01 to 2020.09.30. 87/2020A-N079. Total budget / funding to BU: \$11,775
- NASA New Frontiers Data Analysis Program *The plasma distribution in the Io torus during the Juno epoch*, Co-I (PI Paul Withers, Boston Univ.), 2019.03.21 to 2022.02.28. 80NSSC19K0818. Total budget / funding to BU: \$289,272
- SOFIA Guest Observer Cycle 7 *Io's Atomic Sulfur Atmosphere in the Mid-IR*, PI, 2019.04.01 to 2020.03.31. 07-0221. Total budget / funding to BU: \$16,700
- NASA/NEExSCI NN-EXPLORE WIYN PI Data Award *Confirming a High Velocity Exo-Exosphere at HD 80606b*, PI, 2019.02.01 to 2021.01.31. N0223. Total budget / funding to BU: \$10,100
- NASA/NEExSCI Keck Award *Juno Support: Io's Auroral Emissions in Jovian Eclipse*, PI, 2019.02.01 to 2020.01.31. 84-208B-N110. Total budget / funding to BU: \$10,062
- Hubble Space Telescope Cycle 26 *Auroral and magnetospheric context for Juno in situ instruments during Cycle 26*, Co-I (PI Denis Grodent, Univ. Liege), 2019.03.01 to 2020.02.28. HST-GO-15638. Total budget / funding to BU: \$134,087
- NASA Solar System Workings *Physical Processes Governing Mercury's Alkali Exosphere*, PI, 2018.11.01 to 2021.03.31. 80NSSC19K0790. Total budget: \$352,275. Funding to BU: \$203,872
- NASA Solar System Observations *Ground-based observations of Mercury's exosphere in the post-MESSENGER era*, PI, 2018.03.01 to 2021.02.28. 80NSSC18K0857. Total budget: \$507,403. Funding to BU: \$165,281

- NASA Solar System Workings *The Ins and Outs of the Io Plasma Torus: understanding mass and energy transport using two decades of optical and radio observations*, Co-I (PI Jeff Morgenthaler, Planetary Science Institute), 2017.08.23 to 2020.08.22. 80NSSC17K0733. Total budget: \$526,604. Funding to BU: \$115,358
- Hubble Space Telescope Cycle 25 *Extreme Doppler Shifting of Io's Neutral Jets*, PI, 2018.03.01 to 2019.02.28. HST-GO-15147. Total budget: \$39,999. Funding to BU: \$28,006
- NSF Astronomy and Astrophysics Research Grant *The Influence of Mercury's Magnetosphere on Its Outermost Atmosphere*, Science PI (PI Luke Moore, Boston Univ.), 2016.07.15 to 2019.06.30. AST-1614903. Total budget / funding to BU: \$374,407
- NASA Earth and Space Sciences Fellowship *Mercury's Escaping Atmosphere*, Science PI (PI Michael Mendillo, Boston Univ.), 2010.03.15 to 2013.03.15. 10-Planet10F-0041. Total budget / funding to BU: \$90,000

## Telescope Time Awarded

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|-------------------------------------------------------------|------------|
| JWST, STScI (Co-I, PI Q. Zhang)                             | 2026       |
| Keck II, NASA NExScI (PI)                                   | 2026       |
| JWST, STScI (Co-I, PI Z. Milby)                             | 2025       |
| IRTF, NASA (Co-I, PI R. Cartwright)                         | 2024       |
| Hubble Space Telescope, STScI (Co-I, PI L. Roth)            | 2024       |
| Keck I & II (Co-I, PIs P. Lierle and L. Moore)              | 2024, 2025 |
| HST & JWST joint program, STScI (Co-I, PI K. Retherford)    | 2023       |
| WIYN, NASA NExScI (PI)                                      | 2022       |
| Hubble Space Telescope, STScI (US PI, PI L. Roth)           | 2022       |
| Keck I & II, NASA NExScI (PI & Co-I, PI L. Moore)           | 2022, 2023 |
| Keck I & II, NASA NExScI (Co-I, PIs L. Moore & K. de Kleer) | 2021       |
| Very Large Telescope, ESO (Co-I, PI A. Oza)                 | 2020       |
| Keck I & II, NASA NExScI (PI & Co-I, PI M. Vogt)            | 2020       |
| THEMIS Solar Telescope, SOLARNET (Co-I, PI V. Mangano)      | 2019, 2020 |
| Big Bear Solar Observatory (PI)                             | 2019       |
| GREGOR Solar Telescope (PI)                                 | 2019       |
| SOFIA, USRA (PI)                                            | 2019       |
| IRTF, NASA (Co-I, PI L. Moore)                              | 2019       |
| WIYN, NASA NExScI (PI)                                      | 2019       |
| Keck I, NASA NExScI (PI & Co-I, PI K. de Kleer)             | 2019       |
| Hubble Space Telescope, STScI (Co-I, PI D. Grodent)         | 2019       |
| Hubble Space Telescope, STScI (PI)                          | 2018       |
| Dunn Solar Telescope, National Solar Observatory (PI)       | 2016       |
| Vacuum Tube Telescope, SOLARNET (PI)                        | 2016       |
| GREGOR Solar Telescope, SOLARNET (Co-I, PI V. Mangano)      | 2016       |
| Very Large Telescope, ESO (Co-I, PI B. Bonfond)             | 2015       |

## Teaching

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- AS851 *Graduate Literature Seminar II* (10 students, Spring 2026)
- AS865 & AS866 *Graduate Research Seminar I & II* (~15 students. Fall & Spring 2022 - 2025)
- CC111 *Core Natural Science I: Origins- of the Big Bang, Earth, Life and Humanity* (96 students. Fall 2022)
- AS101 *The Solar System* (~16 students, Summer 2018 - 2021)

## Service (Academic)

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PhD Dissertation Committees:

- Abigail Tadlock (PhD candidate, Boston University) *TBD*
- Lou Baya Ould Rouis (PhD candidate, Boston University) *TBD*
- Alec Daly (PhD 2025, Boston University) *Unraveling Jupiter's Magnetospheric Dynamics: Characteristics and Impacts of Interchange and Injection Events*
- Rozenn Robidel (PhD 2023, Université Paris-Saclay) *Study of the exosphere of Mercury with the PHEBUS spectrograph on the BepiColombo mission*

## Service (Professional)

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- Organizing Committee of the IAU G5 Commission on Stellar & Planetary Atmospheres (2024 - 2027)
- Private sector advisor to the Group of Friends on Dark and Quiet Skies for the UN Committee on the Peaceful Uses of Outer Space (2024)
- Institutional Representative: Massachusetts Space Grant Consortium (2020 - )
- Journal Reviews: Planetary Science Journal, Icarus (outstanding reviewer award), Journal of Geophysical Research, Geophysical Research Letters, Nature, Astronomy & Astrophysics
- Panelist for federal programs: 12x Research Opportunities in Space and Earth Science (ROSES) programs, Keck Time Allocation Committee, PDS Derived Data Review, Discovery Mission Extension Review, CNRS DIM-ORIGINES (France).
- Session Chair: AGU: Dynamics of the Io-Jupiter System (2014) and Dynamic Exospheres of Terrestrial bodies through the Solar System (2023-2025), DPS/EPSC: Fire and Ice: Io and Beyond (2023), Io plasma torus splinter meetings at MOP (2017 & 2018), Exosphere/Magnetosphere, Mercury: Current and Future Science of the Innermost Planet, USRA (2018)

## Team Activities & Memberships

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- Rapid Imaging Planetary Spectrograph, ground-based instrument, PI
- ESA/JAXA BepiColombo mission, Co-I, PHEBUS instrument
- NASA’s Commercial Lunar Payload Services, Co-I, Lunar Environment Heliospheric X-ray Imager
- NASA SBIR Phase I & II to LambdaMetrics for development of the Serpentine Integrated Grating Spectrometer for Extreme Precision Radial Velocimetry, private consultant.
- International Space Science Institute Teams: The influence of Io on Jupiter’s Magnetosphere (2016 - 2017), Surface Bounded Exospheres and Interactions in the Solar System (2020), Mass loss from Io’s unique atmosphere: Do volcanoes really control Jupiter’s magnetosphere? (2021 - 2022)
- Keck SCALES solar system science definition team
- Memberships: American Astronomical Society, International Astronomical Union, American Geophysical Union

## Public Outreach

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|-----------------------------------------------------------------------------------------------------------|-------------|
| Subject Matter Expert, NASA Community College Network                                                     | 2023 –      |
| Volunteer Astronomer, High Rock Tower & Observatory, City of Lynn, MA                                     | 2022 –      |
| Spectroscopy Lab Instructor, ISS Downlink Day, Boston University                                          | 2022        |
| Lab Instructor, BU Academy Dept Visit, Boston Univ.                                                       | 2022        |
| Host, Navajo-Hopi Astronomy Outreach Program, Lowell Observatory                                          | 2018        |
| Lecturer / Volunteer Astronomer, J.B. Coit Observatory, Boston Univ.                                      | 2017 –      |
| Lecturer, Fan Mountain Observatory Public Nights, Univ. Virginia                                          | 2014 – 2015 |
| Science Fair Judge, Virginia Piedmont Regionals, Charlottesville, VA                                      | 2014        |
| Lecturer, McCormick Observatory Public Nights, Univ. Virginia                                             | 2013 – 2015 |
| Lab Instructor, Upward Bound program, Boston Univ.                                                        | 2010        |
| Science Fair Judge, O’Bryant School for Math and Science, Roxbury, MA                                     | 2009        |
| Workshop Coordinator, Sprout, <a href="http://www.thesprouts.org">www.thesprouts.org</a> , Somerville, MA | 2009 – 2013 |

## Press & Media

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|------------------------------------------------------------------------------------------------------|------|
| NY Times: <a href="#">Auroras Spotted on Neptune for the First Time, Lead to a New Mystery</a>       | 2025 |
| Astronomy Magazine: <a href="#">Scientists view Ganymede’s aurorae dancing during eclipse</a>        | 2024 |
| NY Times: <a href="#">Europa, Thought to Be Habitable, May Be Oxygen-Starved</a>                     | 2024 |
| Astronomy Magazine: <a href="#">Aurorae throughout our solar system and beyond</a>                   | 2023 |
| Live TV Interview, Hopewell Meteorite, Russian Television International (RTVi)                       | 2023 |
| AAS NOVA <a href="#">Pizza Aurorae: Northern (and/or Southern) Lights on Io</a>                      | 2023 |
| NY Times: <a href="#">New Auroras Found Glowing in the Skies of Jupiter’s Moons</a>                  | 2023 |
| NY Times: <a href="#">Telescopes Team Up to Forecast an Alien Storm on Titan</a>                     | 2022 |
| Boston Globe: <a href="#">Bad weather may hurt viewing of rare lunar eclipse Friday in Mass.</a>     | 2021 |
| Swedish National Public Television: <a href="#">Today a storm from the moon pulls past the earth</a> | 2021 |
| NY Times: <a href="#">The Moon Has a Comet-Like Tail</a>                                             | 2021 |

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|-------------------------------------------------------------------------------------------|-------------|
| Wall Street Journal: <i>Comet Neowise as Seen Around the World</i>                        | 2020        |
| Sky & Telescope: <i>Comet NEOWISE Dazzles at Dusk</i>                                     | 2020        |
| Fox News: <i>Comet NEOWISE may have sodium tail, new images suggest</i>                   | 2020        |
| TV Interview, <i>Space Challenges Documentary</i> , Bulgarian National Public Television  | 2017        |
| TV Interview, NASA ScienceCast: <i>The 2016 Transit of Mercury</i>                        | 2016        |
| Content Advisor, Science in the News, Harvard University GSAS                             | 2013 - 2016 |
| Radio Interview, Science Straight Up, WTJU FM                                             | 2014        |
| Phys.org: <i>Mercury's comet-like appearance spotted by satellites looking at the Sun</i> | 2010        |
| Universe Today: <i>STEREO Catches Mercury Acting Like a Comet</i>                         | 2010        |